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Develop a Customized Ping Command to test the Server connectivity

Aim: To test the Server connectivity and measure network performance using the ping command.

Uses OF Ping:

* Ping is a network diagnostic command used to check whether a server or Host is reachable over a network.

1. What does ping Measure?

Reachability - checks whether the destination server can be contacted

Latency - Measures the time taken for packets to travel to the server and back (in milliseconds)

Packet loss - shows whether any packets were lost during transmission, indicating Possible network problems.

1. When is ping Useful?

Ping is commonly Used to

* Verify that a Server is online
* Test network connectivity before Troubleshooting applications
* Measure network response time
* detect packet loss caused by unstable connections
* Perform basic network health checks

Procedure:

1. Basic ping command:

Ping www.google.com

Output: Displays IP address of the server, no of Packets sent, received, lost & the Avg round-trip time

2 Ping localHost Command:

Ping localhost

Output: Replies from 127.0.0.1 Confirm that the TCP/IP stack on local System is working Properly.

3. Ping Using IP Address command:

Ping 8.8.8.8

Output: Displays connectivity & delay time (in milliseconds) blw your system and Google DNS server

4. continuos ping Test command:

Ping www.google.com -t

Output: continuosly pings the server until stopped manually Showing live latency & packet information

5. ping with specific number of Echo requests command:

Ping www.google.com -n 5

Output: Sends exactly 5 echo request & displays the result with avg time & packet loss summary

6. ping with packet size Specification:

Ping www.google.com -l 1000

output: sends a ping request with Packet Size of 1000 bytes, showing transmission performance

Result: The ping commands were Successfully executed in windows command prompt & the network connectivity, latency & packet transmission blw local comp's & various servers were verified